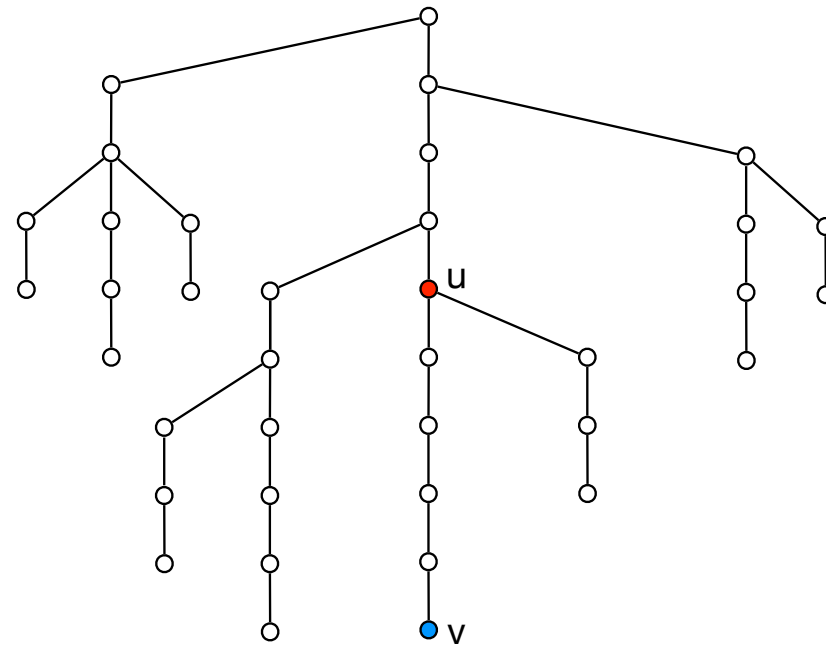


Level Ancestor

Philip Bille/Inge Li Gørtz

Level Ancestor

- **Level ancestor problem.** Preprocess rooted tree T with n nodes to support
 - $LA(v,k)$: return the k th ancestor of node v .



$$LA(v, 5) = u$$

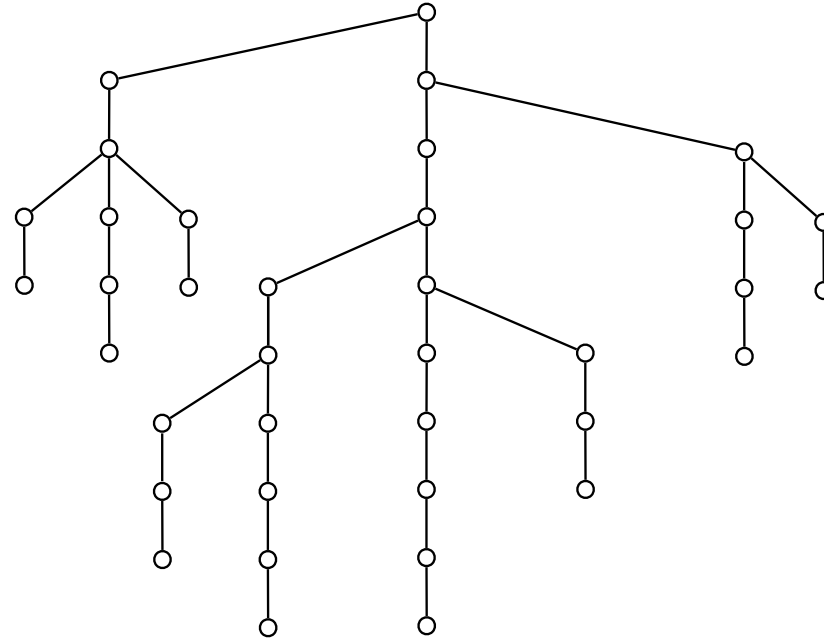
Level Ancestor

- [Applications.](#)
 - Basic primitive for navigating trees (any hierarchical data).
 - Illustration of wealth of techniques for trees.
 - Path decompositions.
 - Tree decomposition.
 - Tree encoding and tabulation.

Level Ancestor

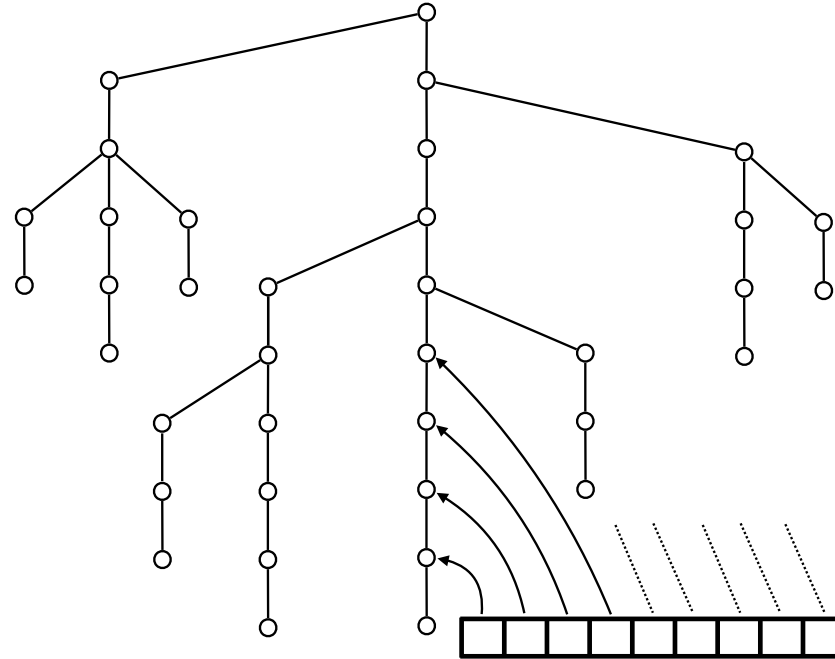
- **Goal.** Linear space and constant time.
- **Solution in 7 steps (!).**
 - **No data structure.** Very slow, little space
 - **Direct shortcuts.** Very fast, lot of space.
 -
 - **Ladder decomposition + jump pointers + top-bottom decomposition.** Very fast, little space.

Solution 1: No Data Structure



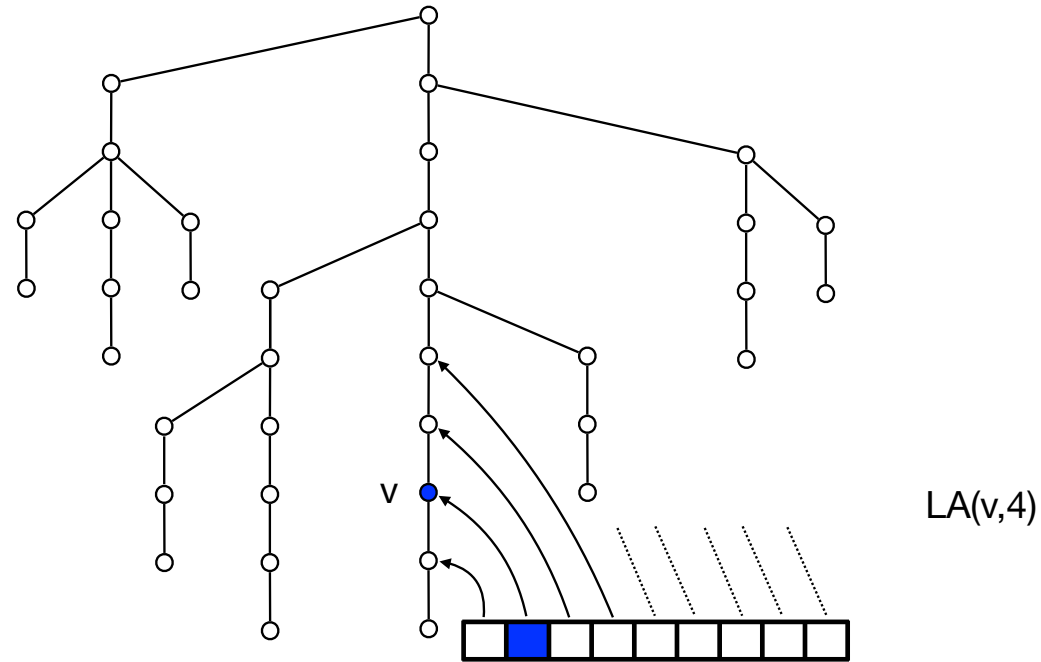
- **Data structure.** Store tree T (using pointers).
- **LA(v,k):** Walk up.
- **Time.** $O(n)$
- **Space.** $O(n)$

Solution 2: Direct Shortcuts



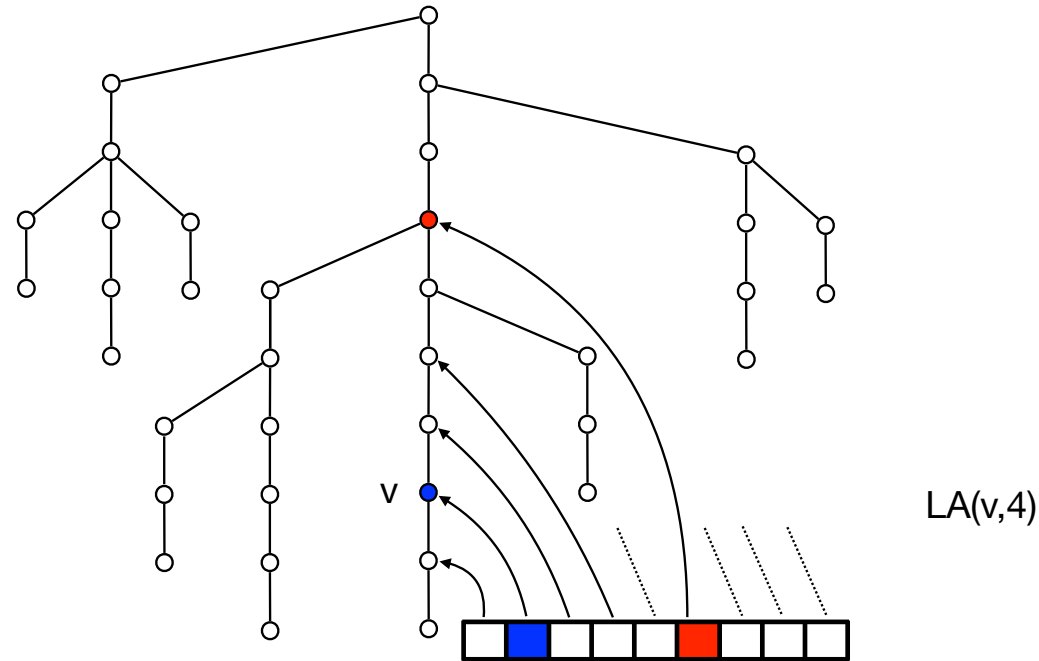
- **Data structure.** Store each root-to-leaf path in an array.
- $LA(v,k)$: **Jump** up.

Solution 2: Direct Shortcuts



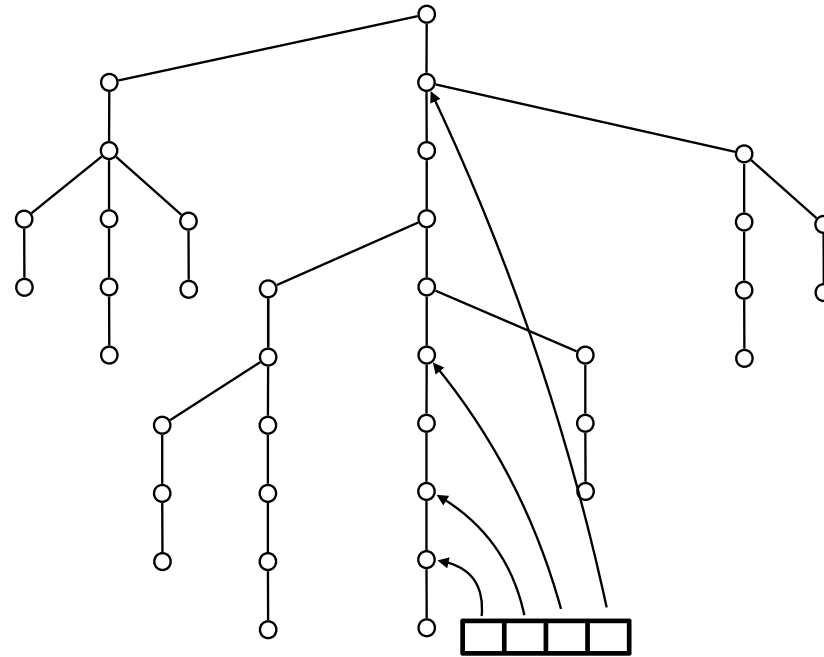
- **Data structure.** Store each root-to-leaf path in an array.
- $LA(v, k)$: **Jump** up.

Solution 2: Direct Shortcuts



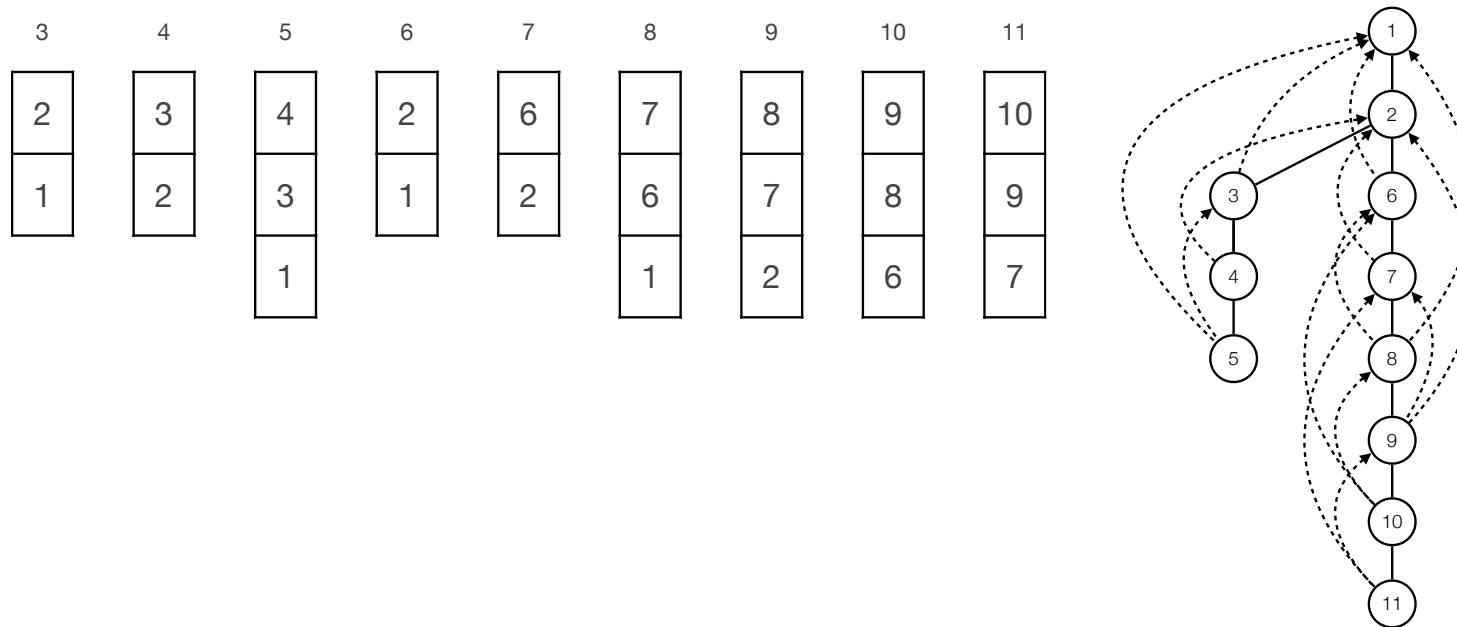
- **Data structure.** Store each root-to-leaf path in an array.
- $LA(v,k)$: **Jump** up.
- **Time.** $O(1)$
- **Space.** $O(n^2)$

Solution 3: Jump Pointers



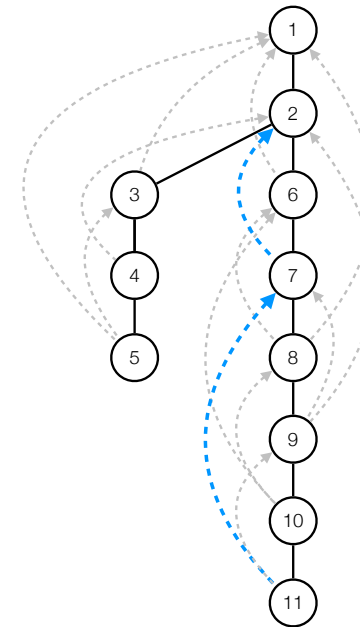
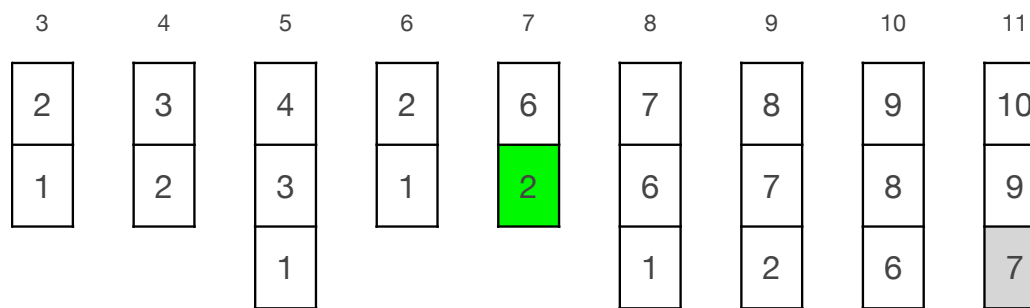
- **Data structure.** For each node v , store pointers to ancestors at distance 1, 2, 4, ..
- **LA(v, k):** Jump to most distant ancestor no further away than k . Repeat.
- **Time.** $O(\log n)$
- **Space.** $O(n \log n)$

Solution 3: Jump Pointers



- **Data structure.** For each node v , store pointers to ancestors at distance 1, 2, 4, ..
- **LA(v, k):** Jump to most distant ancestor no further away than k . Repeat.
- **Time.** $O(\log n)$
- **Space.** $O(n \log n)$

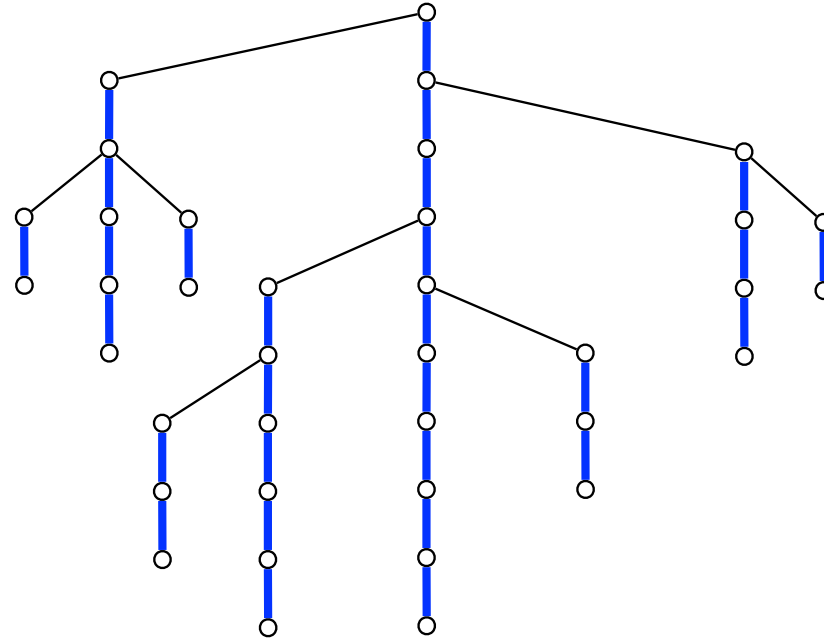
Solution 3: Jump Pointers



$$LA(11,6) = LA(7,2) = 2$$

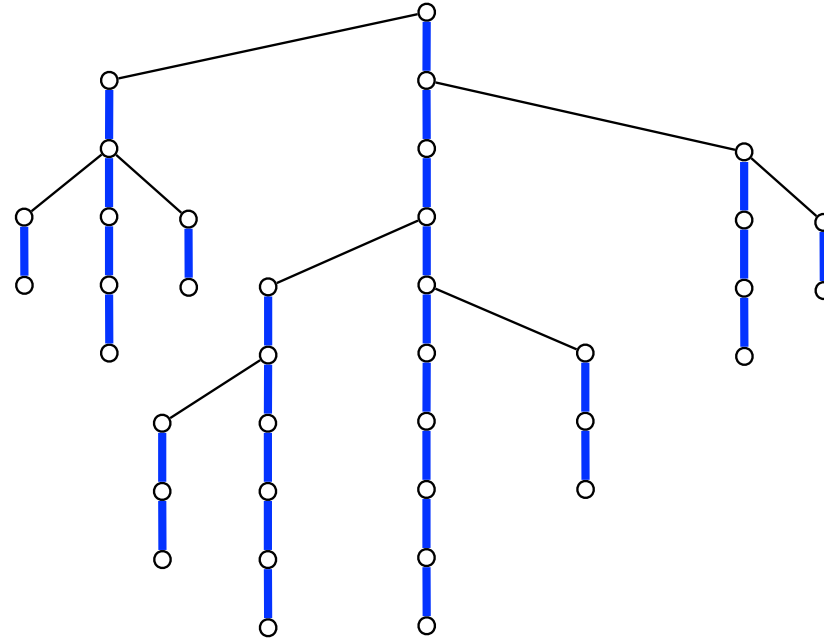
- **Data structure.** For each node v , store pointers to ancestors at distance 1,2,4, ..
- **$LA(v,k)$:** Jump to most distant ancestor no further away than k . Repeat.
- **Time.** $O(\log n)$
- **Space.** $O(n \log n)$

Solution 4: Long Path Decomposition



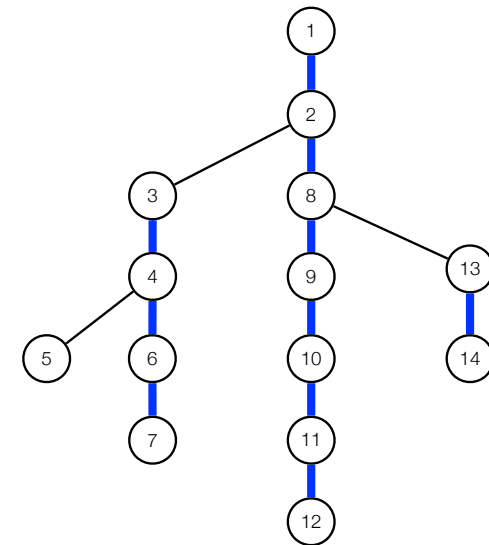
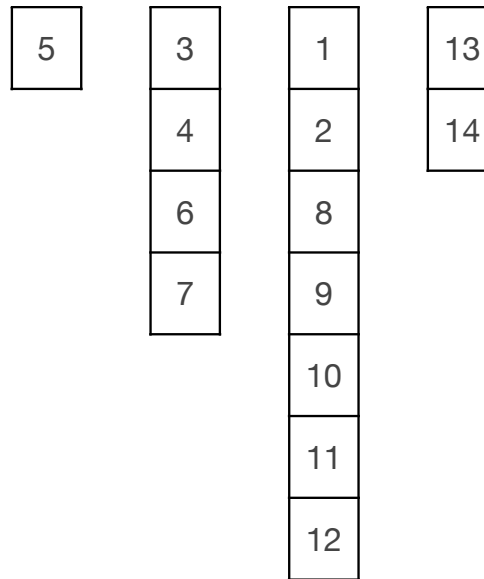
- Long path decomposition.
 - Find root-to-leaf path p of maximum length.
 - Recursively apply to subtrees hanging off p .
- Lemma. Any root-to-leaf path passes through at most $O(n^{1/2})$ long paths.
- Longest paths partition $T \Rightarrow$ total length (number of nodes) of all longest paths is $= n$

Solution 4: Long Path Decomposition



- **Data structure.** Store each long path in array.
- **LA(v,k):** Jump to kth ancestor or root of long path. Repeat.
- **Time.** $O(n^{1/2})$
- **Space.** $O(n)$

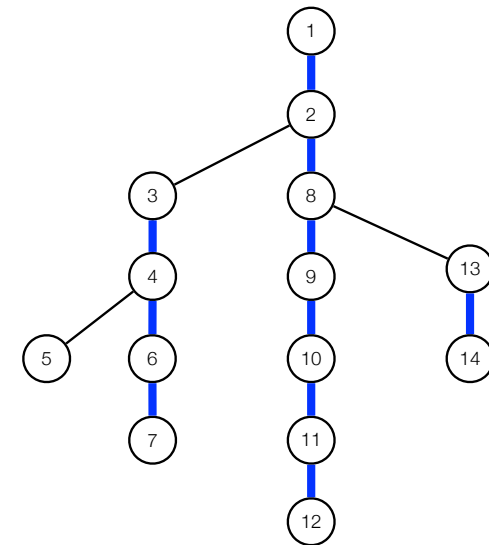
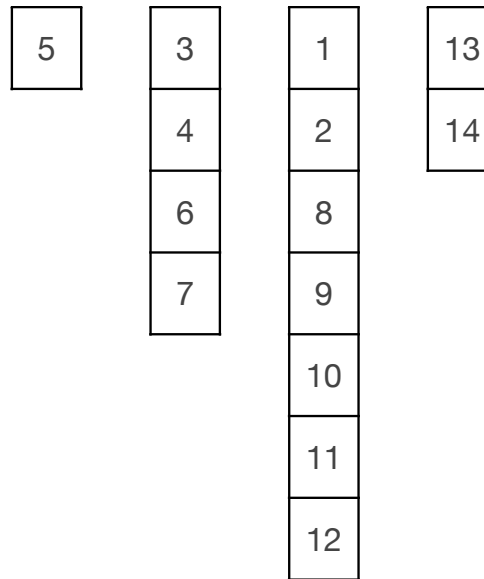
Solution 4: Long Path Decomposition



- **Data structure.** Store each long path in array.
- **LA(v,k):** Jump to kth ancestor or root of long path. Repeat.
- **Time.** $O(n^{1/2})$
- **Space.** $O(n)$

Solution 4: Long Path Decomposition

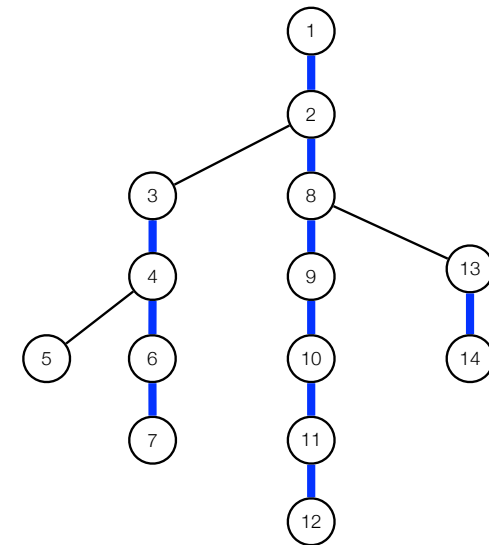
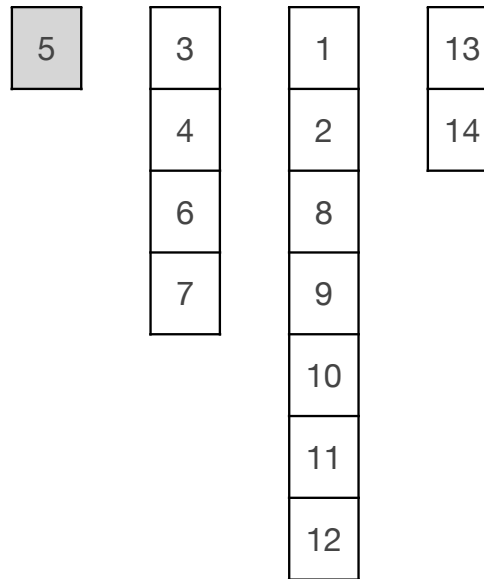
LA(5,4)



- **Data structure.** Store each long path in array.
- **LA(v,k):** Jump to kth ancestor or root of long path. Repeat.
- **Time.** $O(n^{1/2})$
- **Space.** $O(n)$

Solution 4: Long Path Decomposition

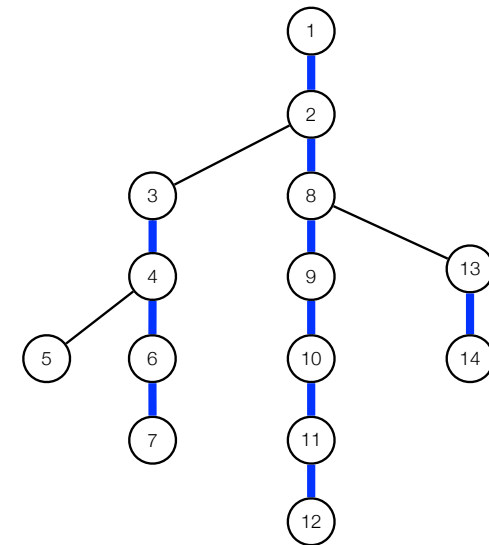
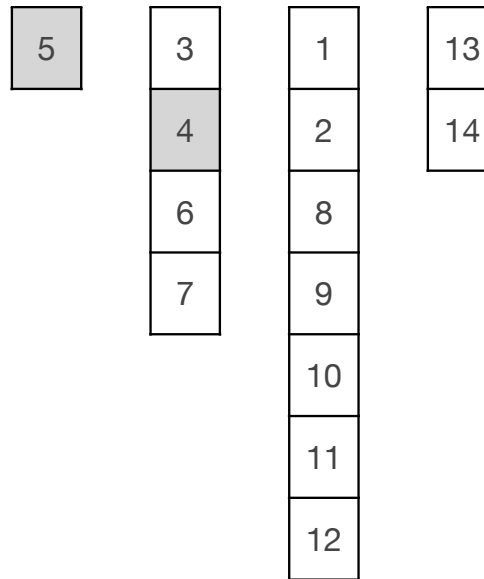
LA(5,4)



- **Data structure.** Store each long path in array.
- **LA(v,k):** Jump to kth ancestor or root of long path. Repeat.
- **Time.** $O(n^{1/2})$
- **Space.** $O(n)$

Solution 4: Long Path Decomposition

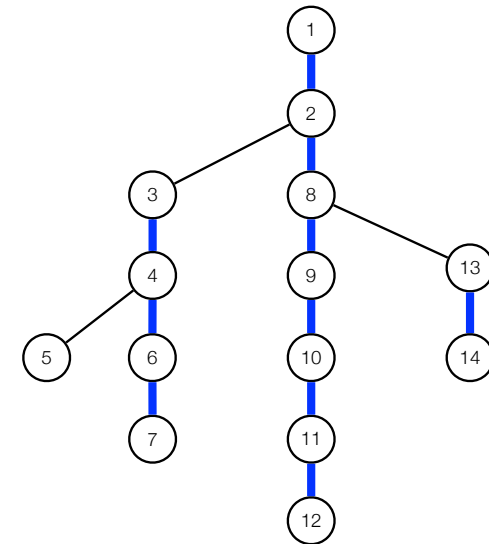
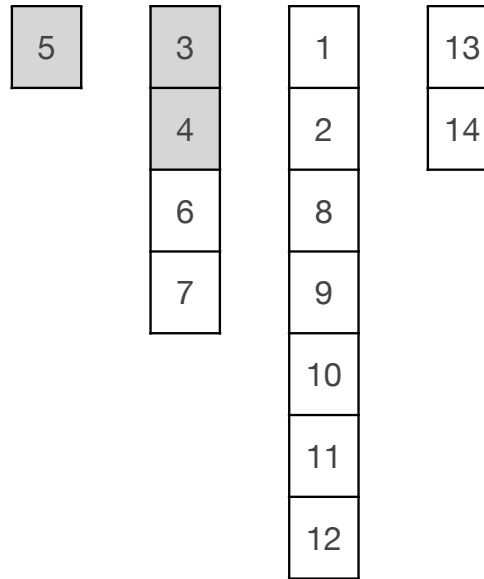
$$LA(5,4) = LA(4,3)$$



- **Data structure.** Store each long path in array.
- **$LA(v,k)$:** Jump to k th ancestor or root of long path. Repeat.
- **Time.** $O(n^{1/2})$
- **Space.** $O(n)$

Solution 4: Long Path Decomposition

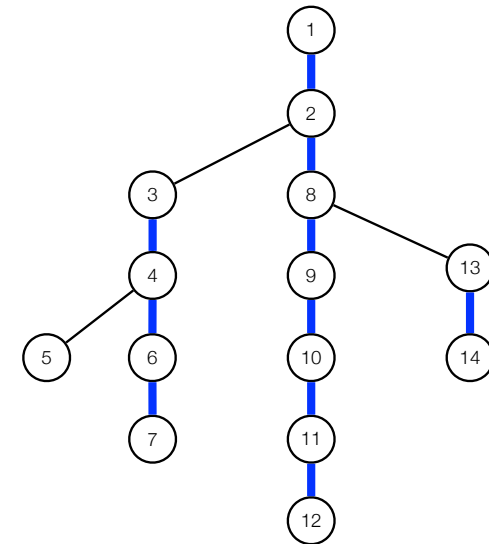
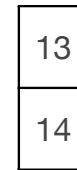
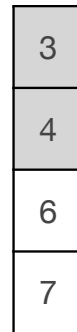
$$LA(5,4) = LA(4,3) = LA(3,2)$$



- **Data structure.** Store each long path in array.
- **$LA(v,k)$:** Jump to k th ancestor or root of long path. Repeat.
- **Time.** $O(n^{1/2})$
- **Space.** $O(n)$

Solution 4: Long Path Decomposition

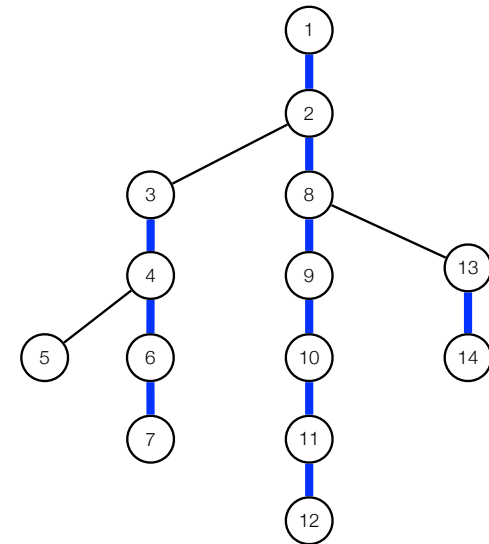
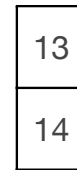
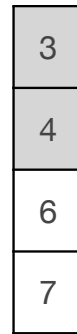
$$LA(5,4) = LA(4,3) = LA(3,2) = LA(2,1)$$



- **Data structure.** Store each long path in array.
- **$LA(v,k)$:** Jump to k th ancestor or root of long path. Repeat.
- **Time.** $O(n^{1/2})$
- **Space.** $O(n)$

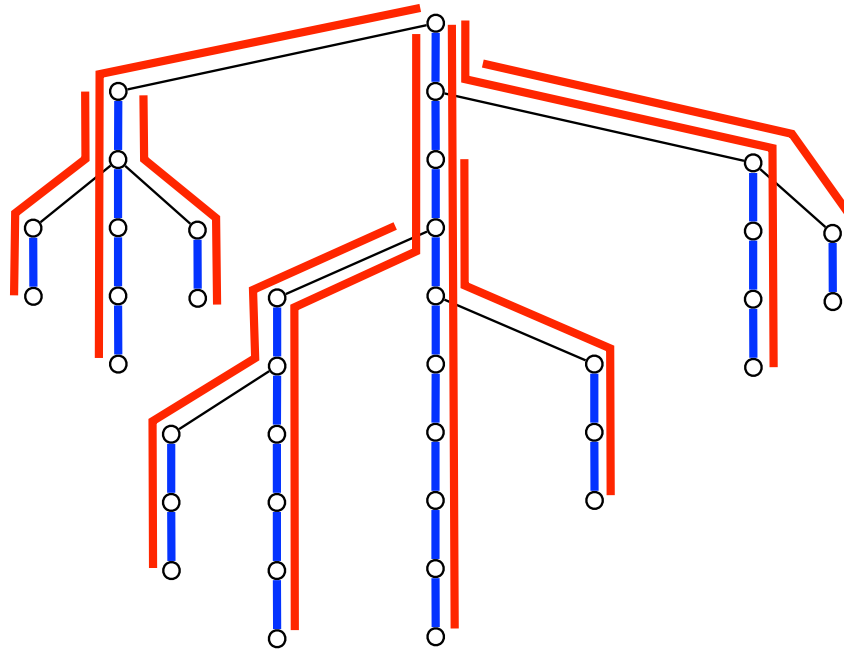
Solution 4: Long Path Decomposition

$$LA(5,4) = LA(4,4) = LA(3,3) = LA(2,1) = 1$$



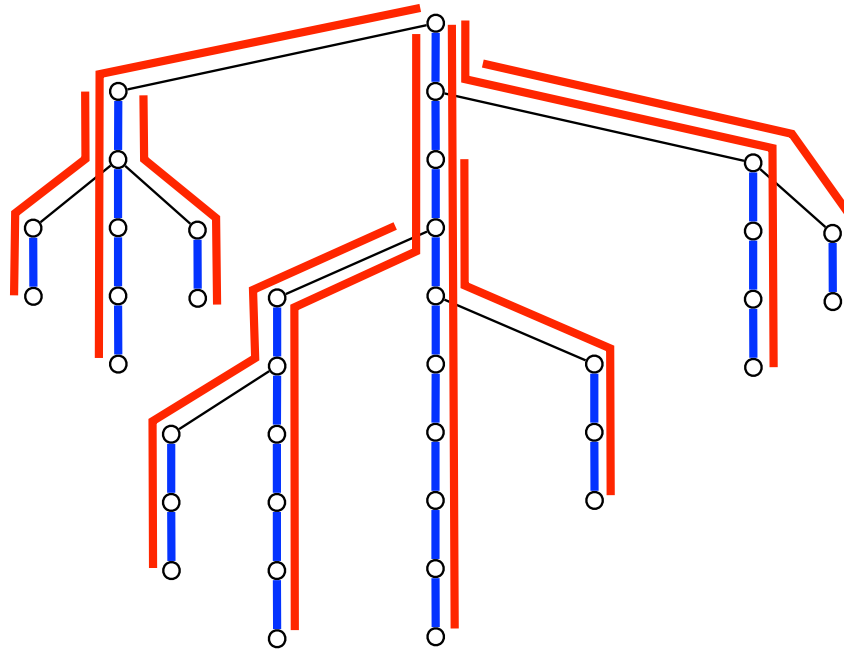
- **Data structure.** Store each long path in array.
- **$LA(v,k)$:** Jump to k th ancestor or root of long path. Repeat.
- **Time.** $O(n^{1/2})$
- **Space.** $O(n)$

Solution 5: Ladder Decomposition



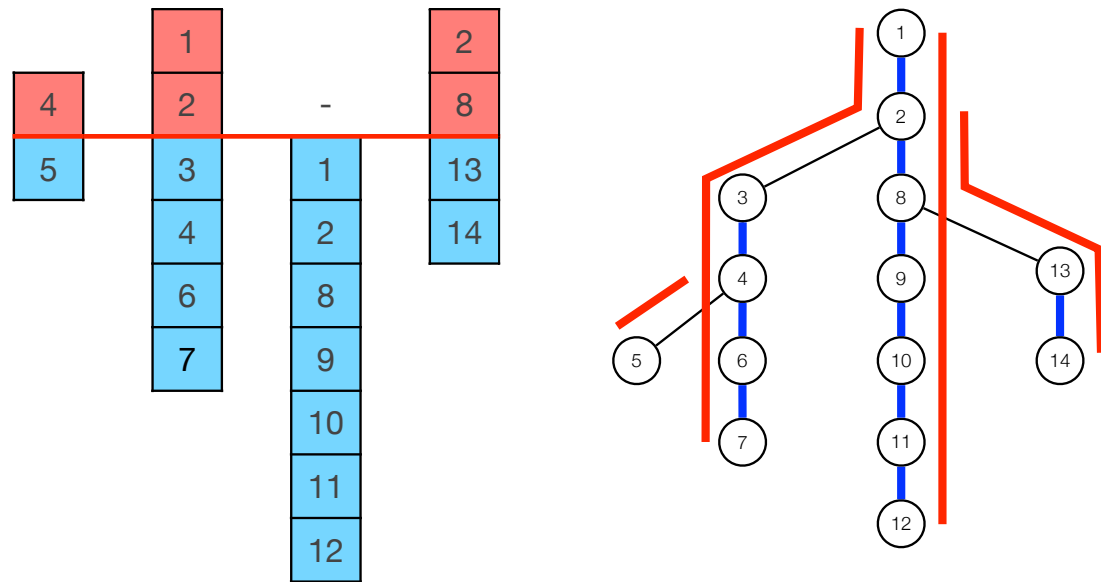
- Ladder decomposition.
 - Compute long path decomposition.
 - Double each long path.
- Lemma. Any root-to-leaf path passes through at most $O(\log n)$ ladders.
- Total length of ladders is $\leq 2n$.

Solution 5: Ladder Decomposition



- **Data structure.**
 - Store each ladder in array.
 - Each node points to ladder corresponding to its longest path.
- **LA(v,k):** Jump to kth ancestor or root of ladder. Repeat.
- **Time.** $O(\log n)$
- **Space.** $O(n)$

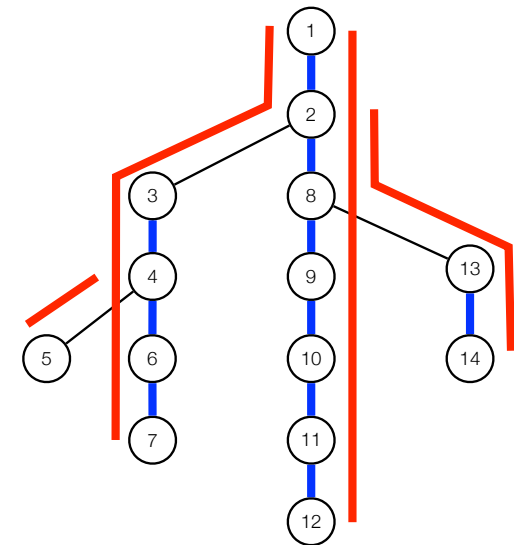
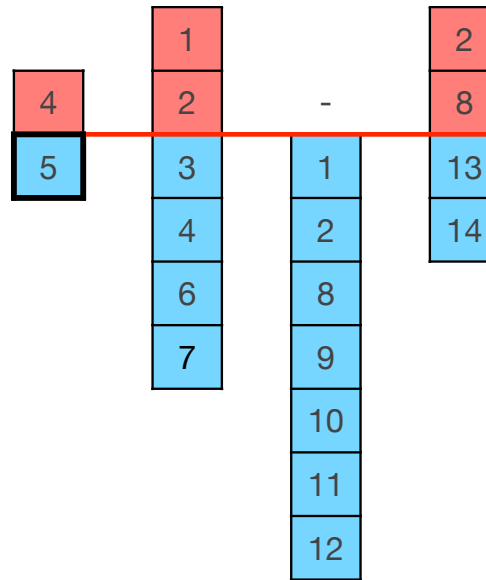
Solution 5: Ladder Decomposition



- **Data structure.**
 - Store each ladder in array.
 - Each node points to ladder corresponding to its longest path.
- **LA(v,k):** Jump to kth ancestor or root of ladder. Repeat.
- **Time.** $O(\log n)$
- **Space.** $O(n)$

Solution 5: Ladder Decomposition

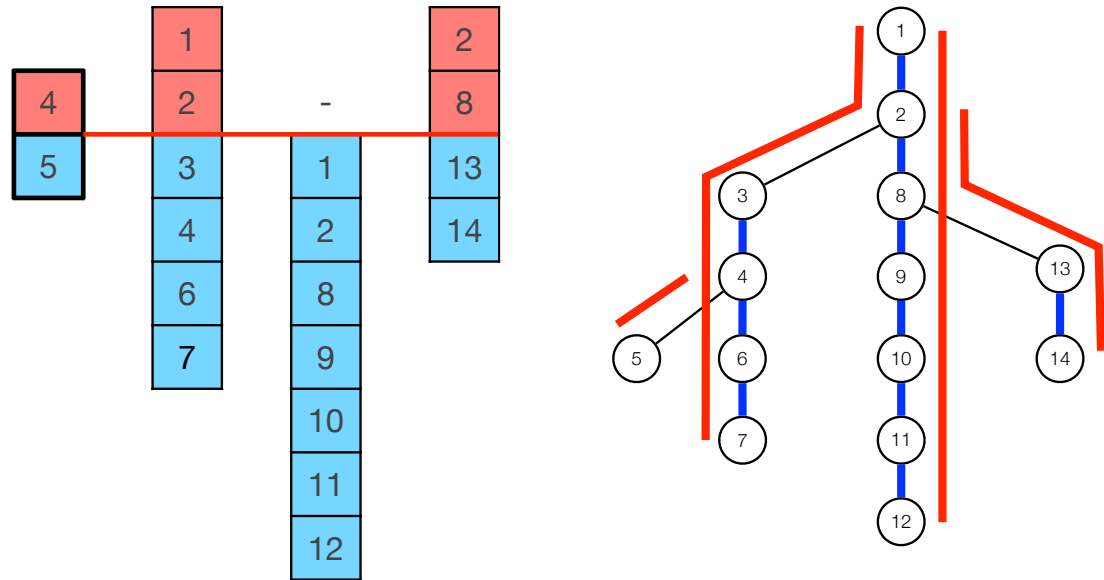
LA(5,4)



- **Data structure.**
 - Store each ladder in array.
 - Each node points to ladder corresponding to its longest path.
- **LA(v,k):** Jump to kth ancestor or root of ladder. Repeat.
- **Time.** $O(\log n)$
- **Space.** $O(n)$

Solution 5: Ladder Decomposition

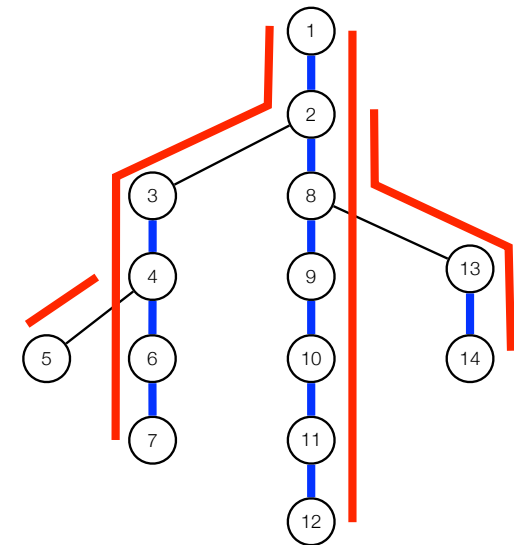
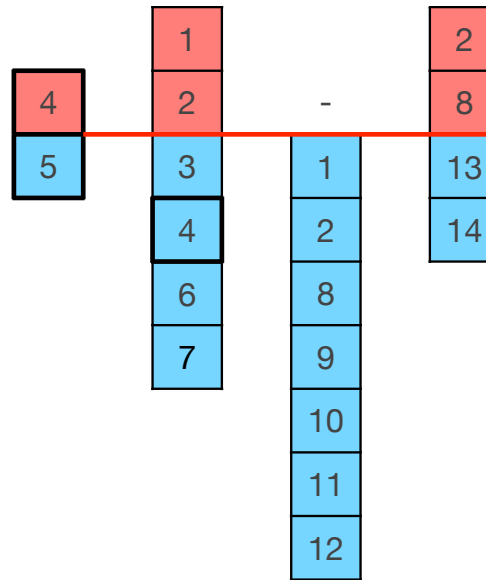
$$LA(5,4) = LA(4,3)$$



- **Data structure.**
 - Store each ladder in array.
 - Each node points to ladder corresponding to its longest path.
- **$LA(v,k)$:** Jump to k th ancestor or root of ladder. Repeat.
- **Time.** $O(\log n)$
- **Space.** $O(n)$

Solution 5: Ladder Decomposition

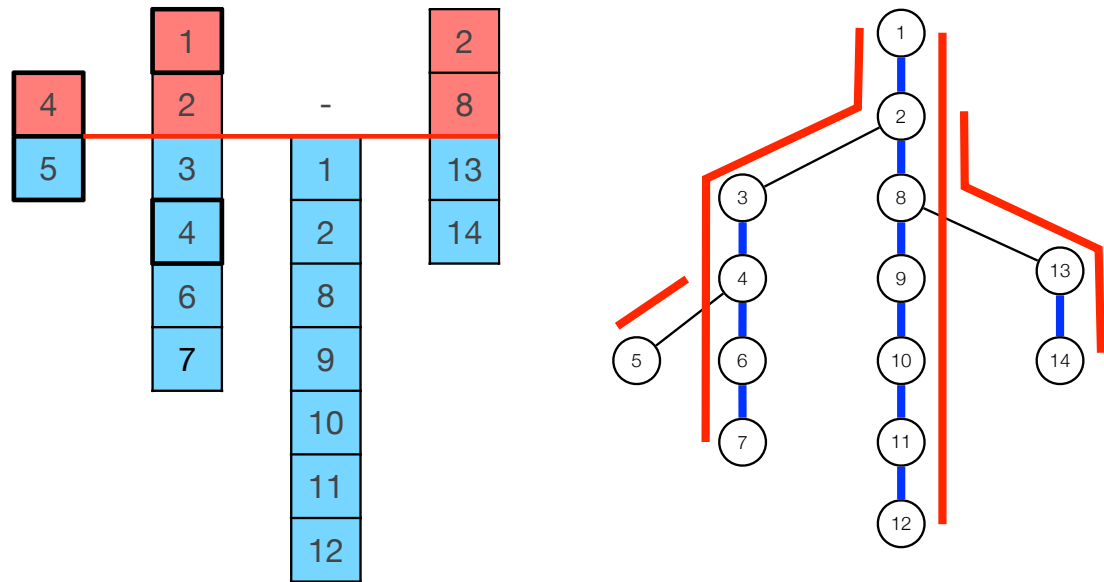
$$LA(5,4) = LA(4,3)$$



- **Data structure.**
 - Store each ladder in array.
 - Each node points to ladder corresponding to its longest path.
- **LA(v,k):** Jump to kth ancestor or root of ladder. Repeat.
- **Time.** $O(\log n)$
- **Space.** $O(n)$

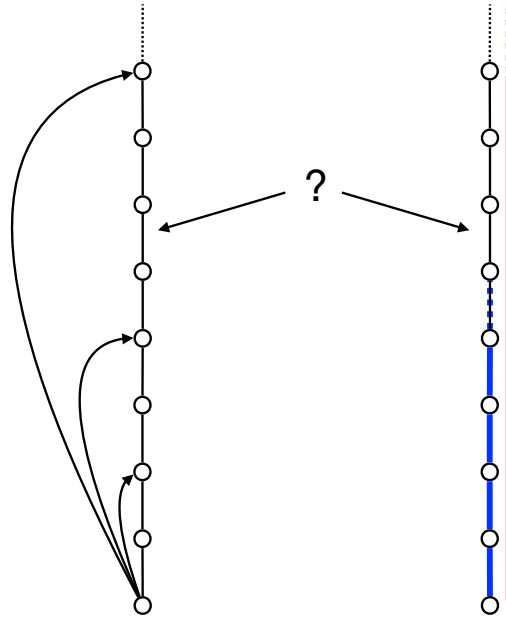
Solution 5: Ladder Decomposition

$$LA(5,4) = LA(4,3) = 1$$



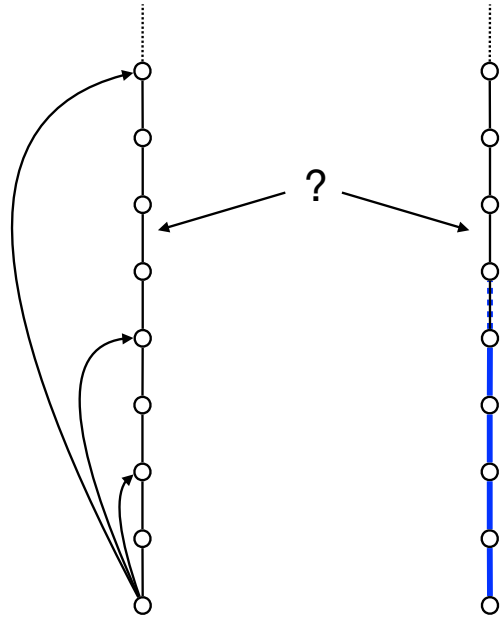
- **Data structure.**
 - Store each ladder in array.
 - Each node points to ladder corresponding to its longest path.
- **LA(v,k):** Jump to kth ancestor or root of ladder. Repeat.
- **Time.** $O(\log n)$
- **Space.** $O(n)$

Solution 6: Ladder Decomposition + Jump Pointers



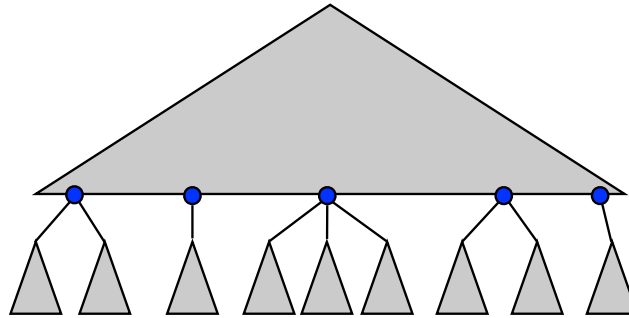
- **Data structure.** Ladder decomposition + Jump pointers.
- **LA(v,k):**
 - Jump to most distant ancestor not further away than k using jump pointer.
 - Jump to kth ancestor using ladder.
- **Time.** $O(1)$
- **Space.** $O(n) + O(n \log n) = O(n \log n)$

Solution 6: Ladder Decomposition + Jump Pointers



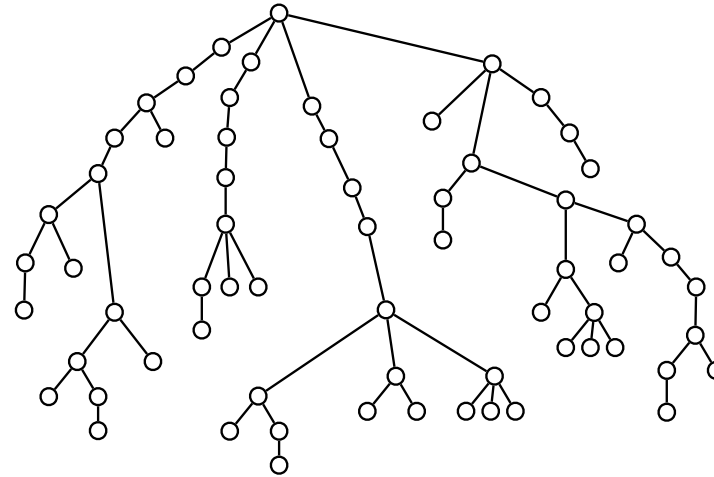
- **Correctness.**
 - A node at height x is on a ladder of height at least $2x$.
 - After jump we are at a node of height at least $k/2$.
 - $\Rightarrow$ after jump we are at a ladder that contains our goal.

Solution 7: Top-Bottom Decomposition



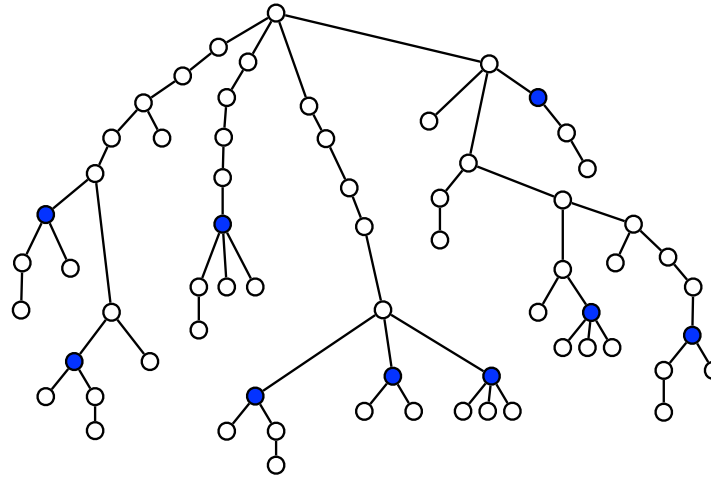
- **Jump nodes.** Maximal **deep** nodes with $\geq 1/4 \log n$ descendants.
- **Top tree.** Jump nodes + ancestors.
- **Bottom trees.** Below top tree.

Solution 7: Top-Bottom Decomposition



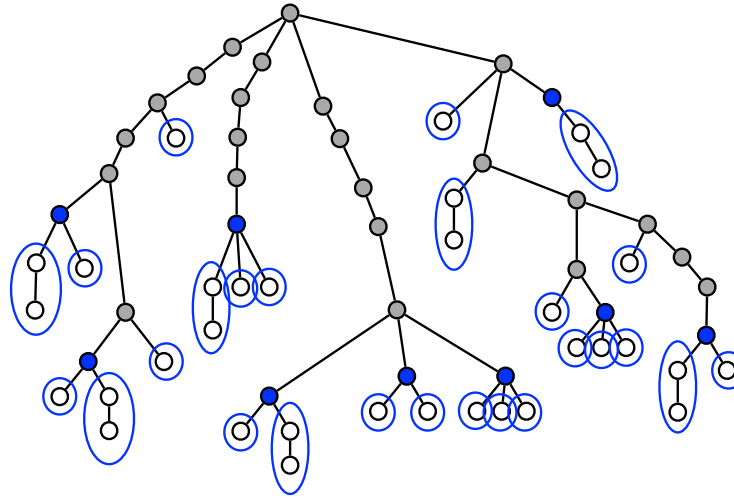
- **Jump nodes.** Maximal **deep** nodes with $\geq \frac{1}{4} \log n$ descendants.
- **Top tree.** Jump nodes + ancestors.
- **Bottom trees.** Below top tree.

Solution 7: Top-Bottom Decomposition



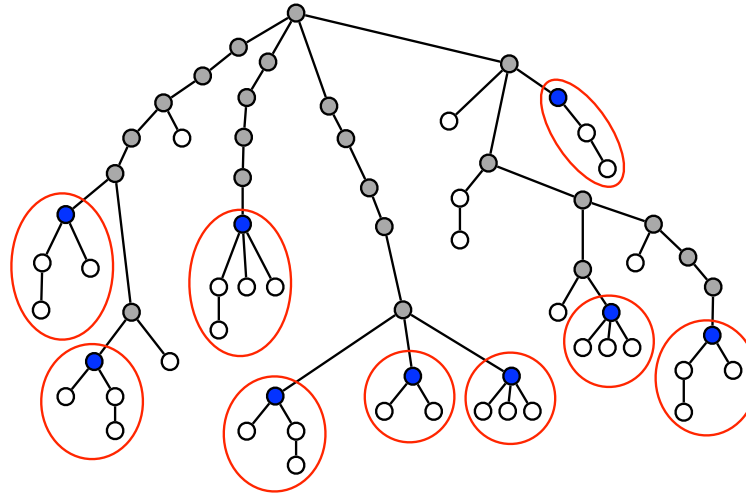
- **Jump nodes.** Maximal **deep** nodes with $\geq 1/4 \log n$ descendants.
- **Top tree.** Jump nodes + ancestors.
- **Bottom trees.** Below top tree.

Solution 7: Top-Bottom Decomposition



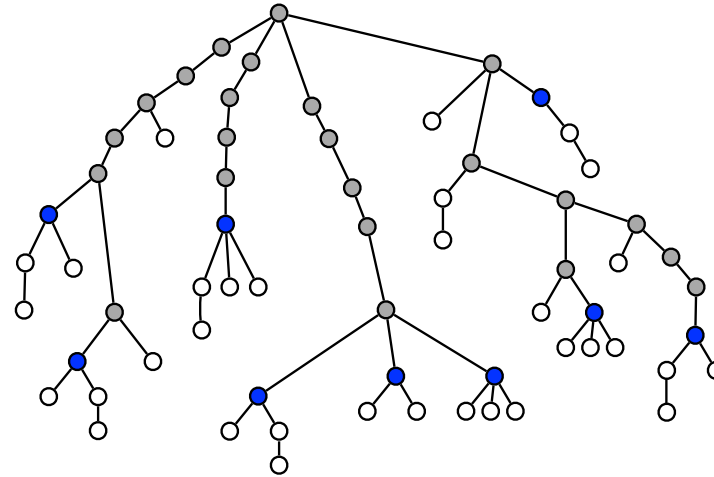
- **Jump nodes.** Maximal **deep** nodes with $\geq 1/4 \log n$ descendants.
- **Top tree.** Jump nodes + ancestors.
- **Bottom trees.** Below top tree.
- Size of each bottom tree $< 1/4 \log n$.

Solution 7: Top-Bottom Decomposition



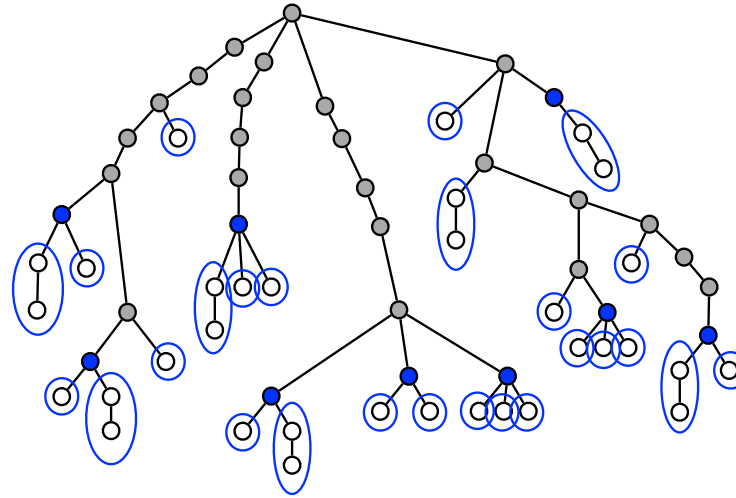
- **Jump nodes.** Maximal **deep** nodes with $\geq 1/4 \log n$ descendants.
- **Top tree.** Jump nodes + ancestors.
- **Bottom trees.** Below top tree.
- Size of each bottom tree $< 1/4 \log n$.
- Number of jump nodes is at most $O(n/\log n)$.

Solution 7: Top-Bottom Decomposition



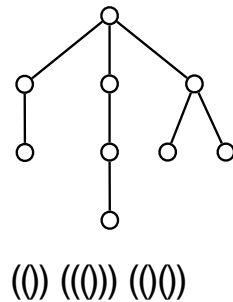
- Data structure for top.
 - Ladder decomposition + Jump pointers for jump nodes.
 - For each internal node pointer to some jump node below.
- $LA(v,k)$ in top:
 - Follow pointer to jump node below v .
 - Jump pointer + ladder solution.
- Time. $O(1)$
- Space. $O(n) + (n/\log n \cdot \log n) = O(n)$

Solution 7: Top-Bottom Decomposition



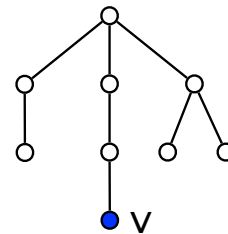
- Data structure for the bottom trees.
 - Tabulation.

Solution 7: Top-Bottom Decomposition



- **Tree encoding.** Encode each bottom tree B using balanced parentheses representation.
 - $< 2 \cdot \frac{1}{4} \log n = \frac{1}{2} \log n$ bits.

Solution 7: Top-Bottom Decomposition



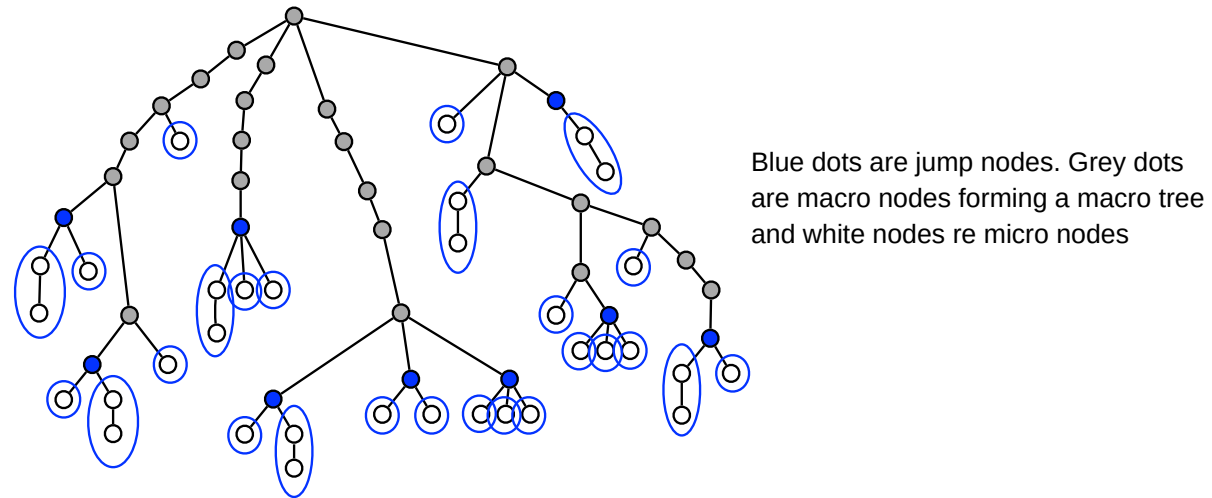
v has preorder number 5 = 0101_2

(()) ((())) (())

$\text{Code}(B, v, 2) = 0011000111001011 \text{ } 0101 \text{ } 0010$

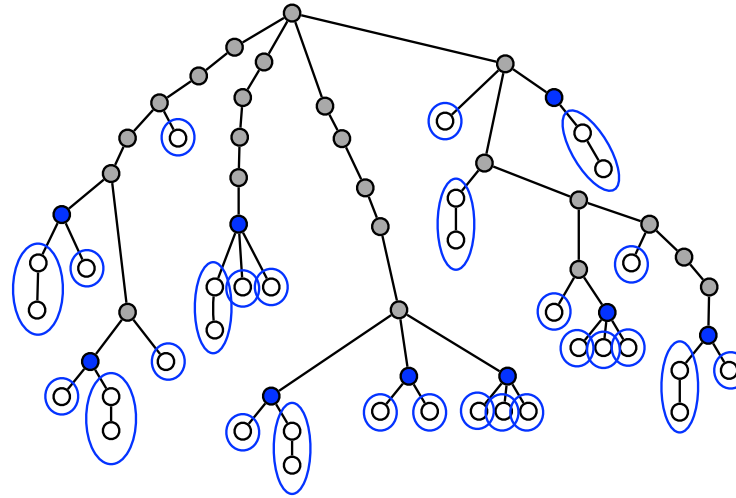
- **Tree encoding.** Encode each bottom tree B using balanced parentheses representation.
 - $< 2 \cdot 1/4 \log n = 1/2 \log n$ bits.
- **Integer encoding.** Encode inputs v and k to LA
 - $< 2 \cdot \log(1/4 \log n) < 2 \log \log n$ bits.
- **LA encoding.** Concatenate into $\text{code}(B, v, k)$
 - $\Rightarrow |\text{code}(B, v, k)| < 1/2 \log n + 2 \log \log n$ bits.

Solution 7: Top-Bottom Decomposition



- **Data structure for bottom.**
 - Build table A s.t. $A[\text{code}(B, v, k)] = \text{LA}(v, k)$ in bottom tree B.
- **LA(v,k) in bottom:** Lookup in A.
- **Time.** $O(1)$
- **Space.** $2^{|\text{code}|} < 2^{1/2 \log n + 2 \log \log n} = n^{1/2} \log^2 n = o(n)$.

Solution 7: Top-Bottom Decomposition



- Query in bottom with answer in top?
 - For nodes in bottom trees store root of bottom tree and distance to this.
- Combine bottom and top data structures $\Rightarrow O(n)$ space and $O(1)$ query time.

Solution 7: Top-Bottom Decomposition

- **Theorem.** We can solve the level ancestor problem in linear space and constant query time.